

Fluid Dynamics

1 Kinematics

1.1 Streamlines and pathlines

There are two natural ways to think of flow. We will use the Euclidean perspective throughout this course. We think of ourselves as stationary observers, watching the flow go past, and we can measure the velocity field as $\mathbf{u}(\mathbf{x}, t)$. There is also the Lagrangian perspective, being a moving observer travelling along with the flow, but this is more difficult.

Definition 1.1 – Streamlines

Streamlines are curves that are everywhere parallel to the flow at a given instant. The streamline that goes through $\mathbf{x}_0$ at time t_0 is given parametrically by $\mathbf{x} = \mathbf{x}(s, \mathbf{x}_0, t_0)$ satisfying

$$\frac{d\mathbf{x}}{ds} = \mathbf{u}(\mathbf{x}, t_0)$$

with $\mathbf{x} = \mathbf{x}_0$ at $s = 0$.

The set of streamlines shows the direction of flow at a given instant in time.

Definition 1.2 – Pathlines

Pathlines (also called Lagrangian trajectories) describe the trajectory of a fluid particle (a very small bit of fluid). The pathline $\mathbf{x} = \mathbf{x}(t, \mathbf{x}_0)$ of fluid which is at $\mathbf{x}_0$ at $t = 0$ satisfies

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}, t)$$

with $\mathbf{x} = \mathbf{x}_0$ at $t = 0$.

The streamlines show all fluid particles at one time while the pathlines show one fluid particle at all times.

Example 1.1

A two-dimensional example with $\mathbf{u}(t) = (1, t)$. We have

$$\begin{cases} dx/dt = 1 \\ dy/dt = t \end{cases} \implies \begin{cases} x = x_0 + t \\ y = y_0 + t^2/2 \end{cases}$$

so the pathlines are $y - y_0 = (x - x_0)^2/2$.

If the flow is steady ($\mathbf{u}$ does not depend on t), then the streamlines and pathlines are identical.

1.2 The material derivative

Consider a quantity $F(\mathbf{x}, t)$ in a fluid flow. Suppose we are on a boat travelling with the flow and we are measuring F over time. We would like to compute the rate of change of F , denoted DF/Dt . Let δt be a small time interval, we have

$$\begin{aligned}\delta F &= F(\mathbf{x} + \delta \mathbf{x}, t + \delta t) - F(\mathbf{x}, t) \\ &= \delta t \frac{\partial F}{\partial t} + (\delta \mathbf{x} \cdot \nabla)F + \text{higher order terms.}\end{aligned}$$

Since we assume that the observer is moving with the flow, $\delta \mathbf{x} = \mathbf{u} \delta t$. Hence,

Definition 1.3 – Material derivative

$$\frac{DF}{Dt} = \frac{\partial F}{\partial t} + (\mathbf{u} \cdot \nabla)F$$

$\partial F / \partial t$ is the local derivative and $(\mathbf{u} \cdot \nabla)F$ is the advected derivative.

1.3 Conservation of mass

Consider flow through a straight rigid pipe with a constant cross section. It is possible for the fluid flowing out to be faster than the fluid flowing in? For gases, it might be. But for liquids, it is not possible since they are incompressible.

Define the mass density as $\rho(\mathbf{x}, t)$. Consider a fixed volume V and compute the rate of change of its mass $M = \int_V \rho dV$. The mass can only change due to flow of mass across the boundary surface ∂V . Let δA be a small surface element and $\mathbf{n}$ be its normal vector pointing outward. The volume of fluid moving out of V during the time interval δt is $(\mathbf{u} \cdot \mathbf{n}) \delta A \delta t$. Hence,

$$\begin{aligned}\frac{dM}{dt} &= - \int_{\partial V} \rho (\mathbf{u} \cdot \mathbf{n}) \delta A \\ &= - \int_V \nabla \cdot (\rho \mathbf{u}) dV.\end{aligned}$$

Since this is true for all V , we have

$$\frac{d\rho}{dt} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

Using $\nabla \cdot (\rho \mathbf{u}) = \rho \nabla \cdot \mathbf{u} + \mathbf{u} \cdot \nabla \rho$, we can rewrite this as

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u}.$$

In this course, we assume that fluid is incompressible so the density of a fluid particle cannot change and

$$\frac{D\rho}{Dt} = 0 \iff \nabla \cdot \mathbf{u} = 0.$$

Also, we will assume that ρ is constant.

1.4 Kinematic boundary condition

Suppose that the material boundary of a body of fluid has velocity $\mathbf{U}(\mathbf{x}, t)$. At a point $\mathbf{x}$ on the boundary, the fluid velocity relative to the surface is $\mathbf{u} - \mathbf{U}$. The amount of mass crossing the interface over a small surface

element δA during a time interval δt is

$$\begin{aligned}\rho(\mathbf{u} - \mathbf{U}) \cdot \mathbf{n} \delta A \delta t &= 0 \\ \mathbf{u} \cdot \mathbf{n} &= \mathbf{U} \cdot \mathbf{n}\end{aligned}$$

on the interface. We can think of this as mass conservation at a boundary. Since $\mathbf{n}$ appears on both sides, there is no need to normalise $\mathbf{n}$.

- (i) If the boundary is fixed, then $\mathbf{U} = \mathbf{0}$ and $\mathbf{u} \cdot \mathbf{n} = 0$.
- (ii) Consider an air-water interface, perhaps on the ocean. Let $z = \zeta(x, y, t)$ be the water height at (x, y) and time t . We can think of the water surface as $F(x, y, z, t) = 0$ with $F(x, y, z, t) = z - \zeta(x, y, t)$. The normal $\mathbf{n}$ is parallel to $\nabla F = (-\zeta_x, -\zeta_y, 1)$. Let $\mathbf{u} = (u, v, w)$. Since $\mathbf{U} = (0, 0, \zeta_t)$, the kinematic boundary condition becomes

$$-u\zeta_x - v\zeta_y + w = \zeta_t \iff w = \zeta_t + u\zeta_x + v\zeta_y = \frac{D\zeta}{Dt} \iff \frac{DF}{Dt} = 0.$$

1.5 Streamfunction for 2D incompressible flow

Recall that incompressible flow satisfies $\nabla \cdot \mathbf{u} = 0$. Hence, there exists a vector field $\mathbf{A}$ such that $\mathbf{u} = \nabla \times \mathbf{A}$. Suppose we are working in 2D Cartesians.

Definition 1.4 – Streamfunction

Let $\mathbf{u} = (u, v, 0)$ and $\mathbf{A} = (0, 0, \psi(x, y))$. We have

$$u = \frac{\partial \psi}{\partial y} \text{ and } v = -\frac{\partial \psi}{\partial x}.$$

We call $\psi(x, y)$ the streamfunction, with dimension $L^2 T^{-1}$.

Example 1.2

Consider $\mathbf{u} = (y, x)$. Note that $\nabla \cdot \mathbf{u} = 0$.

$$\begin{aligned}\frac{\partial \psi}{\partial y} &= u = y \implies \psi = \frac{1}{2}y^2 + f(x) \\ -\frac{\partial \psi}{\partial x} &= -f'(x) = x \implies f = C - \frac{1}{2}x^2\end{aligned}$$

Hence, $\psi = (y^2 - x^2)/2 + C$.

- (i) Streamlines are given by $\psi = \text{constant}$.
- (ii) $|\mathbf{u}| = |\nabla \psi|$ so the flow is faster if the streamlines are closer.
- (iii) Consider two points $\mathbf{x}_0$ and $\mathbf{x}_1$ in 2D. The volume flux crossing the line between $\mathbf{x}_0$ and $\mathbf{x}_1$ is

$$\int_{\mathbf{x}_0}^{\mathbf{x}_1} \mathbf{u} \cdot \mathbf{n} \, dl = \psi(\mathbf{x}_1) - \psi(\mathbf{x}_0).$$

- (iv) ψ is constant at a rigid boundary.

In polar coordinates (r, θ) , we have $\mathbf{u} = (u_r(r, \theta), u_\theta(r, \theta), 0)$ and $\mathbf{A} = \psi(r, \theta)\mathbf{e}_z$.

$$\mathbf{u} = \nabla \times \mathbf{A} = \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta}, -\frac{\partial \psi}{\partial r}, 0 \right).$$

We can also check that $\nabla \cdot \mathbf{u} = 0$.

2 Dynamics of inviscid flow

2.1 Surface and volume forces

There are two types of forces exerted on a fluid.

- (i) Forces that are proportional to the volume. For example, gravity.
- (ii) Forces that are proportional to the surface area. For example, pressure or viscous stresses.

We first consider volume forces. Denote $\mathbf{f}(\mathbf{x}, t) \delta V$ as the force acting on a small volume element δV . Often, we have a conservative $\mathbf{f}$ so $\mathbf{f} = -\nabla \chi$. For gravity, we have $\mathbf{f} = \rho \mathbf{g}$ and $\chi = \rho g z$.

For surface forces, consider a small element of area $\mathbf{n} \delta A$, where $\mathbf{n}$ points from the $-$ side to the $+$ side. Denote the surface force exerted by the $+$ side on the $-$ side by $\boldsymbol{\tau}(\mathbf{x}, t, \mathbf{n}) \delta A$. We call $\boldsymbol{\tau}$ the stress acting on the surface element. Its dimension is F L^{-2} . Note that $\boldsymbol{\tau}$ depends on orientation. By Newton's third law, $\boldsymbol{\tau}(\mathbf{x}, t, -\mathbf{n}) = -\boldsymbol{\tau}(\mathbf{x}, t, \mathbf{n})$.

There are many cases where friction inside the fluid (viscous stress) is negligible. For example, it takes hours for viscosity to bring a litre of water to rest.

Definition 2.1 – Inviscid fluid

A fluid is said to be inviscid if we can neglect viscosity.

For inviscid flows, $\boldsymbol{\tau}$ has no tangential component and its magnitude is independent of orientation. Hence, we can write

$$\boldsymbol{\tau}(\mathbf{x}, t, \mathbf{n}) = -p(\mathbf{x}, t) \mathbf{n}$$

where p is for pressure. The minus sign is there because the $+$ side pushes with pressure p towards the $-$ side when $p > 0$. Note that at the fluid surface, the stress exerted by the fluid on the surface is $p \mathbf{n}$.

2.2 The Euler momentum equation

For the rest of Chapter 2, we would only consider inviscid flow. We will derive an equation similar to mass conservation but with momentum. Consider an arbitrary fixed volume V with boundary ∂V . The momentum inside V is $\int_V \rho \mathbf{u} dV$. It can change due to the flux of momentum across ∂V or forces acting on V and ∂V . The volume flowing out of δA in time δt is $(\mathbf{u} \cdot \mathbf{n}) \delta A \delta t$. So, the momentum flowing out is $\rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) \delta A \delta t$. Hence,

2.1 – Euler momentum integral equation

$$\frac{d}{dt} \int_V \rho \mathbf{u} dV = \underbrace{- \int_{\partial V} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dA}_{\text{momentum through boundary}} + \underbrace{\int_V \mathbf{f} dV}_{\text{volume force}} + \underbrace{\int_{\partial V} -p \mathbf{n} dA}_{\text{surface force}}$$

In components,

$$\int_V \frac{\partial}{\partial t} (\rho u_i) dV = - \int_{\partial V} \rho u_i u_j n_j dA + \int_V f_i dV + \int_{\partial V} -p n_i dA.$$

We call $\rho u_i u_j$ the momentum flow tensor. By the divergence theorem,

$$- \int_{\partial V} \rho u_i u_j n_j dA + \int_{\partial V} -p n_i dA = \int_V \left[-\frac{\partial}{\partial x_j} (\rho u_i u_j) - \frac{\partial p}{\partial x_i} \right] dV.$$

Since this is true for any fixed volume V ,

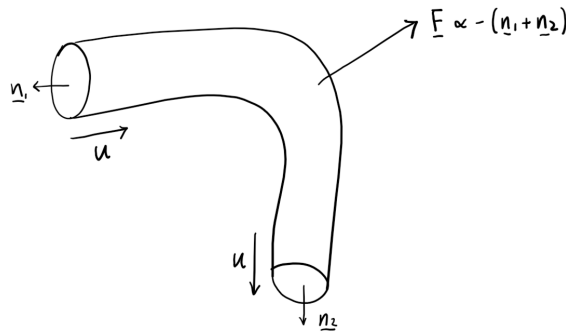
$$\begin{aligned} \frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) &= -\frac{\partial p}{\partial x_i} + f_i \\ u_i \left[\underbrace{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j)}_{= 0 \text{ by mass conservation}} \right] + \rho \left[\frac{\partial u_i}{\partial t} + \underbrace{u_j \frac{\partial}{\partial x_j}}_{(\mathbf{u} \cdot \nabla)} u_i \right] &= -\frac{\partial p}{\partial x_i} + f_i. \end{aligned}$$

2.2 – Euler momentum equation

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f}$$

This is the equation of motion for inviscid fluid flow, an analogue of $m\mathbf{a} = \mathbf{F}$ per unit volume. We have ∇p instead of p in the equation since it is a difference in pressure that moves fluid. In conclusion, fluid particles accelerate under differences in pressure and volume forces.

Example 2.1



Consider fluid flowing through a bent pipe. Assume steady flow and no gravity. What is the force exerted by the flow on the pipe? We can solve with the Euler momentum integral equation. $(d/dt) \int_V \rho \mathbf{u} dV = 0$ since the flow is steady. $\int_V \mathbf{f} dV = 0$ since there is no gravity. Hence,

$$\left(\int_{\text{walls}} + \int_{\text{ends}} \right) [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) + p \mathbf{n}] dA = 0.$$

Note that $\mathbf{u} \cdot \mathbf{n} = 0$ on the walls by the kinematic boundary condition so

$$\int_{\text{walls}} [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) + p \mathbf{n}] dA = \int_{\text{walls}} p \mathbf{n} dA = \text{force exerted by fluid on wall}.$$

Also,

$$\begin{aligned} \int_{\text{ends}} [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) + p \mathbf{n}] dA &= [\rho(-U)(-U\mathbf{n}_1) + p_1 \mathbf{n}_1] A_1 + [\rho U^2 \mathbf{n}_2 + p_2 \mathbf{n}_2] A_2 \\ &= A(p + \rho U^2)(\mathbf{n}_1 + \mathbf{n}_2) \end{aligned}$$

since $p = p_1 = p_2$ as we will see later and we assume that $A = A_1 = A_2$. Hence, the force exerted by the fluid on the wall is $\mathbf{F} = -A(p + \rho U^2)(\mathbf{n}_1 + \mathbf{n}_2)$.

2.3 Bernoulli equation for steady flows with potential forces

We start with the Euler momentum equation

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mathbf{f}.$$

We will assume that the flow is steady so $\partial \mathbf{u} / \partial t = 0$. We will also assume that ρ is constant and $\mathbf{f}$ is a potential force so $\mathbf{f} = -\nabla \chi$. With these assumptions, we have $\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla(p + \chi)$. A vector calculus identity from IA is $(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla(u^2/2) - \mathbf{u} \times (\nabla \times \mathbf{u})$. Let $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ be the vorticity.

$$\begin{aligned} \rho[\nabla(u^2/2) - \mathbf{u} \times (\nabla \times \mathbf{u})] &= -\nabla(p + \chi) \\ \nabla \left[\frac{1}{2} \rho u^2 + p + \chi \right] &= \rho \mathbf{u} \times \boldsymbol{\omega} \end{aligned}$$

Let $H = \rho u^2/2 + p + \chi$ so $\mathbf{u} \cdot \nabla H = \rho \mathbf{u} \cdot (\mathbf{u} \times \boldsymbol{\omega}) = 0$. We arrive at the

2.3 – Steady Bernoulli equation

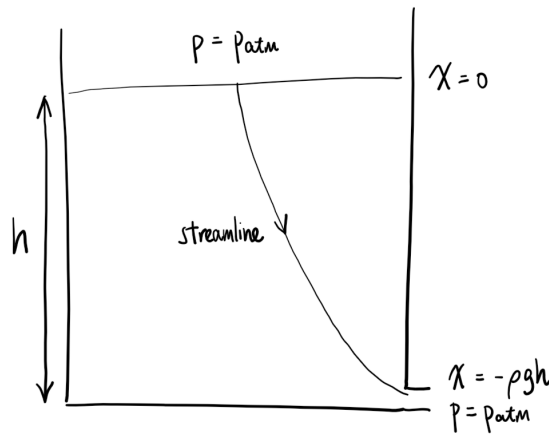
$$\mathbf{u} \cdot \nabla \left[\frac{1}{2} \rho u^2 + p + \chi \right] = 0$$

We see that H is constant along streamlines. A physical consequence is that pressure p decreases when velocity u increases.

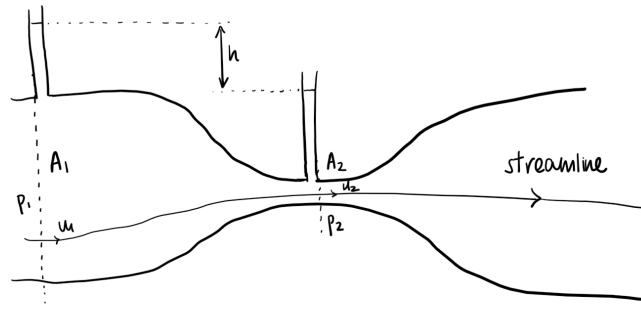
Example 2.2

Suppose we are emptying a tank through a small hole at the bottom. We assume that the hole is small enough so the flow is steady. Apply the steady Bernoulli equation along the streamline to get $H_{\text{top}} = H_{\text{bottom}}$ so

$$\frac{1}{2} \rho u^2 = \rho g h \implies u = \sqrt{2gh}.$$



Example 2.3



Consider a Venturi meter. It is a device to measure flow rate with no moving parts. Ignore gravity. Assume that the flow is steady and is uniform across any cross-section. Let Q be the flow rate so $Q = u_1 A_1 = u_2 A_2$. We want to measure h and infer Q . The steady Bernoulli equation reads

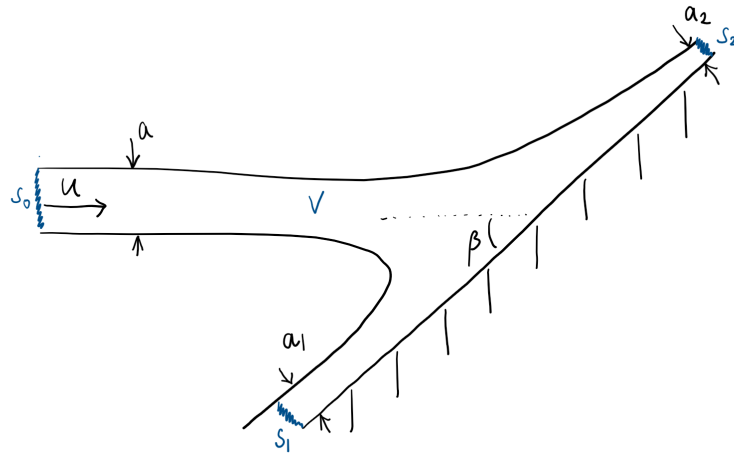
$$\frac{1}{2}\rho u_1^2 + p_1 = \frac{1}{2}\rho u_2^2 + p_2$$

$$p_1 - p_2 = \frac{1}{2}\rho u_1^2 - \frac{1}{2}\rho u_2^2 = \frac{1}{2}\rho u_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right)$$

By hydrostatic balance, $\rho g h = p_1 - p_2$. Simplifying gives

$$Q = \sqrt{2gh} \frac{A_1 A_2}{(A_1^2 - A_2^2)^{1/2}}.$$

Example 2.4



Consider a 2D setup with a water jet hitting an oblique plane. Ignore gravity. Apply Bernoulli to a surface streamline. Since $p = p_{\text{atm}}$ everywhere, we have $u = U$ everywhere. By mass conservation, $aU = a_1U + a_2U$ so $a = a_1 + a_2$. Apply the Euler momentum integral equation to the entire volume of fluid V . We assume steady flow so $(d/dt) \int_V \rho \mathbf{u} dV = 0$. We ignored gravity so $\int_V \mathbf{f} dV = 0$. This gives

$$-\int_{\partial V} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dA + \int_{\partial V} -p \mathbf{n} dA = 0.$$

By the kinematic boundary condition, $\mathbf{u} \cdot \mathbf{n} = 0$ on the surface streamlines and the wall. Also, $p = p_{\text{atm}}$

everywhere except on the wall. Hence,

$$\int_{S_0 \cup S_1 \cup S_2} \rho \mathbf{u} \cdot \mathbf{n} dA + \int_{\text{wall}} p \mathbf{n} dA = 0.$$

The vector component along the wall gives $\rho a U^2 \cos \beta = \rho a_2 U^2 - \rho a_1 U^2$. This implies

$$a_1 = \frac{a(1 - \cos \beta)}{2} \quad a_2 = \frac{a(1 + \cos \beta)}{2}.$$

The component perpendicular to the wall is $F = \int_{\text{wall}} p \mathbf{n} dA = \rho a U^2 \sin \beta$.

2.4 Hydrostatic pressure and Archimedes

Assuming no flow ($\mathbf{u} = \mathbf{0}$) in the Euler momentum equation gives $\mathbf{0} = -\nabla p + \mathbf{f}$. There is still a non-trivial distribution of pressure in the fluid. In the case of gravity, $\mathbf{f} = -\nabla \chi$ where $\chi = \rho g z$, so we have $p + \chi = \text{const}$. This gives rise to the hydrostatic pressure $p = p_0 - \rho g z$.

Consider a submerged body with volume V . What is the force $\mathbf{F}$ exerted by the hydrostatic pressure?

$$\mathbf{F} = - \int_{\partial V} p \mathbf{n} dS = - \int_{\partial V} (p_0 - \rho g z) \mathbf{n} dS = - \int_V \nabla (p_0 - \rho g z) dV = \int_V \rho g \hat{\mathbf{z}} dV = \rho g V \hat{\mathbf{z}}.$$

2.4 – Archimedes' principle

The buoyancy force on a submerged body is the weight of the displaced fluid.

If we consider the Euler momentum equation with only gravity

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \rho \mathbf{g},$$

we could write $p = p_{\text{static}} + p' = p_0 - \rho g z + p'$ where p' is the variation of pressure in the fluid due to its motion, and is called the dynamic pressure. The equation then simplifies to

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p'.$$

2.5 Vorticity

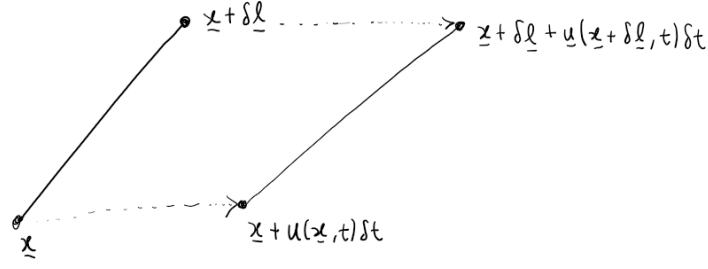
Definition 2.2 – Vorticity

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}$$

Example 2.5

- (i) Consider a solid body rotation with $\mathbf{u} = \boldsymbol{\Omega} \times \mathbf{x}$. The vorticity is $\boldsymbol{\omega} = 2\boldsymbol{\Omega}$.
- (ii) Consider shear flow with $\mathbf{u} = (0, \gamma x, 0)$. The vorticity is $\boldsymbol{\omega} = (0, 0, \gamma)$.
- (iii) Consider a line vortex with $\mathbf{u} = (0, \kappa/(2\pi r), 0)$ in cylindrical coordinates. The vorticity is zero everywhere except at $r = 0$. In fact, $\boldsymbol{\omega} = (0, 0, \kappa \delta(r))$ using Stokes' theorem.

2.5.1 Interpretation of vorticity



Consider a material line $\delta \mathbf{l}$. How does it change from time t to $t + \delta t$? The new $\delta \mathbf{l}$ is

$$\mathbf{x} + \delta \mathbf{l} + \mathbf{u}(\mathbf{x} + \delta \mathbf{l}, t) \delta t - [\mathbf{x} + \mathbf{u}(\mathbf{x}, t) \delta t] = \delta \mathbf{l} + (\delta \mathbf{l} \cdot \nabla) \mathbf{u} \delta t.$$

Hence,

$$\frac{D}{Dt} \delta \mathbf{l} = (\delta \mathbf{l} \cdot \nabla) \mathbf{u}$$

or

$$\frac{D}{Dt} \delta l_i = \left(\delta l_j \frac{\partial}{\partial x_j} \right) u_i$$

in components. We see that the tensor $\nabla \mathbf{u}$ determines changes in material lines and the local rate of deformation. Write

$$\frac{\partial u_i}{\partial x_j} = \underbrace{\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)}_{\text{symmetric } e_{ij}} + \underbrace{\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)}_{\text{antisymmetric}}$$

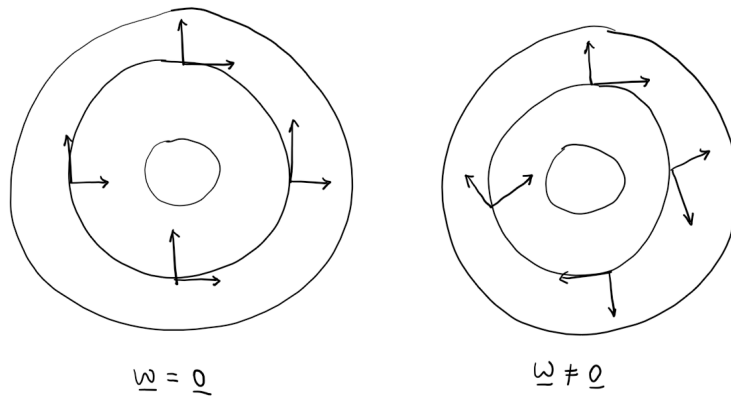
where e_{ij} is the rate of strain tensor measuring structural deformation, and is discussed in Part II. The anti-symmetric part can be written as $\epsilon_{jik} \omega_k / 2$ where $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. What is the contribution of the antisymmetric part to $(D/Dt) \delta l_i$? It is

$$\frac{1}{2} \epsilon_{jik} \omega_k \delta l_j = \frac{1}{2} \epsilon_{ikj} \omega_k \delta l_j = \frac{1}{2} (\boldsymbol{\omega} \times \delta \mathbf{l})_i$$

so

$$\frac{D(\delta \mathbf{l})}{Dt} = \frac{1}{2} \boldsymbol{\omega} \times \delta \mathbf{l} + (\text{deformation}).$$

The first term is describing rigid body rotation with angular velocity $\boldsymbol{\Omega} = \boldsymbol{\omega} / 2$. Hence, $\boldsymbol{\omega}(\mathbf{x}, t) / 2$ represents the average rotation rate of fluid particles. Note that this is not the same as having circular streamlines.



2.5.2 The vorticity equation

We start with the Euler momentum equation and take its curl. Assume that ρ is constant and $\mathbf{f} = -\nabla\chi$. Note that the curl of a gradient is zero and $(\mathbf{u} \cdot \nabla)\mathbf{u} = \nabla(|\mathbf{u}|^2/2) - \mathbf{u} \times \boldsymbol{\omega}$.

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} \right) = -\nabla p - \nabla\chi \implies \frac{\partial}{\partial t}\boldsymbol{\omega} - \nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = \mathbf{0}$$

Look at

$$\begin{aligned} [\nabla \times (\mathbf{u} \times \boldsymbol{\omega})]_i &= \varepsilon_{ijk} \frac{\partial}{\partial x_j} (\varepsilon_{kmn} u_m \omega_n) \\ &= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \left(\frac{\partial u_m}{\partial x_j} \omega_n + \frac{\partial \omega_n}{\partial x_j} u_m \right) \\ &= \omega_j \frac{\partial u_i}{\partial x_j} - \omega_i \frac{\partial u_j}{\partial x_j} + u_i \frac{\partial \omega_j}{\partial x_j} - u_j \frac{\partial \omega_i}{\partial x_j} \\ &\quad \underbrace{\phantom{\omega_j \frac{\partial u_i}{\partial x_j} - \omega_i \frac{\partial u_j}{\partial x_j}}}_{=\nabla \cdot \mathbf{u} = 0} \quad \underbrace{\phantom{u_i \frac{\partial \omega_j}{\partial x_j} - u_j \frac{\partial \omega_i}{\partial x_j}}}_{=\nabla \cdot \boldsymbol{\omega} = 0} \\ &= (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\boldsymbol{\omega}. \end{aligned}$$

Hence,

2.5 – Vorticity equation

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{u} \cdot \nabla)\boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla)\mathbf{u} \qquad \frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla)\mathbf{u}$$

We call $(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}$ the vortex stretching term. Note that $\boldsymbol{\omega}$ only changes if there is a part of $\boldsymbol{\omega}$ that is parallel to $\nabla\mathbf{u}$. In 2D flows, $\boldsymbol{\omega}$ is pointing in the third dimension so it does not change.

2.5.3 Vortex stretching

Compare the evolution equation for material flow and the vorticity equation.

$$\frac{D(\delta\mathbf{l})}{Dt} = \frac{1}{2}\boldsymbol{\omega} \times \delta\mathbf{l} \qquad \frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla)\mathbf{u}$$

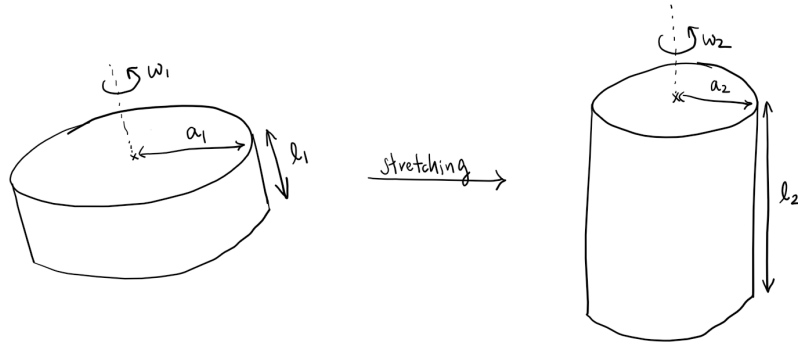
Moving with the fluid, $\boldsymbol{\omega}$ changes just like a material line element initially parallel to $\boldsymbol{\omega}$. In particular, if $\delta\mathbf{l}$ gets longer, then $\boldsymbol{\omega}$ gets larger. This is conservation of angular momentum or the ballerina effect.

Example 2.6

Consider a uniformly rotating fluid. Conservation of mass gives $a_1^2 l_1 = a_2^2 l_2$. Conservation of angular momentum gives $a_1^4 l_1 \omega_1 = a_2^4 l_2 \omega_2$. Hence,

$$\frac{\omega_2}{\omega_1} = \frac{l_2}{l_1}.$$

When the fluid is stretched in the direction of $\boldsymbol{\omega}$, the vorticity increases.



2.5.4 Circulation

Definition 2.3 – Circulation

The circulation around a closed curve Γ is defined as

$$C(t) = \int_{\Gamma} \mathbf{u} \cdot d\mathbf{l} = \iint_S \boldsymbol{\omega} \cdot d\mathbf{S}.$$

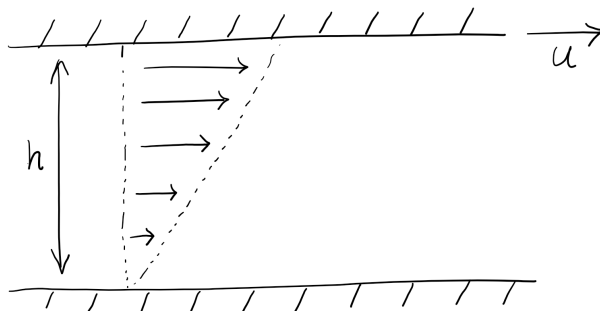
The following is non-examinable. Assume ρ is constant, the fluid is inviscid, and $\mathbf{f}$ is a potential force. Let Γ be a material line. Kelvin's circulation theorem states that $dC/dt = 0$. This implies that if the flow is irrotational at $t = 0$, then it remains irrotational for all time.

3 Introduction to viscous flows

In Chapter 2, we assumed that the fluid was inviscid and neglected any friction in fluid flow. Under these assumptions, we had stress $\boldsymbol{\tau} = -p\mathbf{n}$. Also, we had momentum balance where $\rho D\mathbf{u}/Dt$ depends on ∇p and $\mathbf{f}$. In Chapter 3, we will include viscosity. We will find that there is tangential stress between fluid layers or between the fluid and the boundary. Also, there will be a new term when balancing momentum.

In this course, we will focus on 2D parallel viscous flows. In Cartesian, $\mathbf{u} = (u(y, t), 0, 0)$. Notice in this form, $\nabla \cdot \mathbf{u} = 0$ is automatically satisfied.

3.1 Plane Couette flow and viscosity



Consider a steady flow between two parallel plates, driven only by the motion of the top plate. Experiments have demonstrated that for a large variety of Newtonian fluids (air, water, honey, silicone oil, and so on),

- (i) the fluid velocity is U at the top, and is zero at the bottom;
- (ii) the fluid velocity varies linearly, so $u(y) = Uy/h$;
- (iii) the tangential force per unit area τ required to move the top plate is linear in U and inversely proportional to h , so $\tau \propto U/h$.

Different fluids have a different proportionality constant. We can write

$$\tau = \mu \frac{U}{h}$$

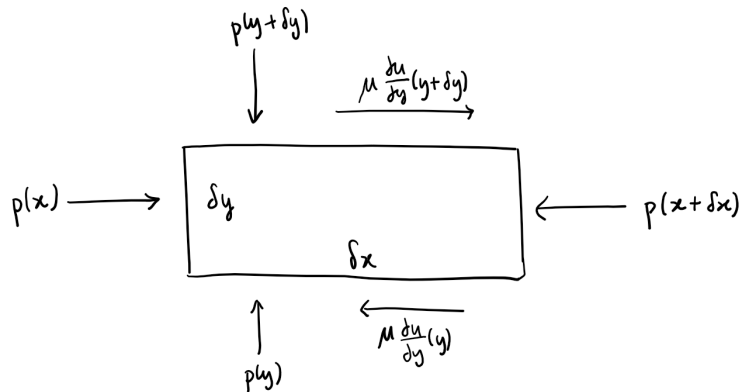
where μ is the dynamic viscosity and is a material property of the fluid. This is Newton's empirical law of viscosity. Locally, the tangential stress exerted by the $+$ side on the $-$ side is

$$\tau = \mu \frac{\partial u}{\partial n}$$

where $\mathbf{n}$ points from the $-$ side to the $+$ side.

3.2 2D parallel viscous flow

3.2.1 Steady case with $\mathbf{f} = 0$



Consider an infinitesimal volume of fluid $\delta x \delta y$. Since the flow is steady, the total force on the volume is zero. In the x -direction,

$$p(x)\delta y - p(x + \delta x)\delta y + \mu \frac{\partial u}{\partial y}(y + \delta y)\delta x - \mu \frac{\partial u}{\partial y}(y)\delta x = 0$$

$$\delta x \delta y \left[-\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} \right] = 0.$$

In the limit,

$$-\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} = 0.$$

In the y -direction,

$$\frac{\partial p}{\partial y} = 0.$$

3.2.2 General case with $u(y, t)$ and $\mathbf{f} \neq 0$

On example sheet 2, we get

$$\rho \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} + f_x \quad 0 = -\frac{\partial p}{\partial y} + f_y$$

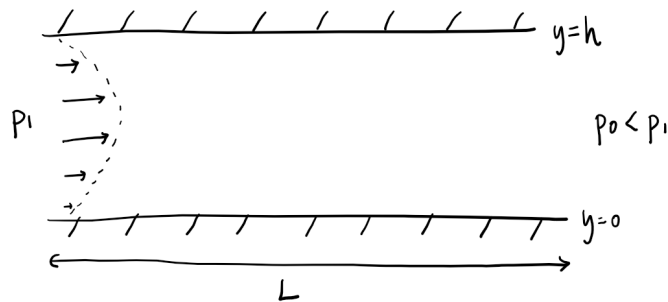
Note that $(\mathbf{u} \cdot \nabla)\mathbf{u} = 0$ in 2D parallel flow, so $\partial u / \partial t$ indeed describes acceleration.

3.2.3 Boundary conditions

Experiments have shown that for viscous flow, $\mathbf{u} = \mathbf{U}$ at a rigid boundary, even down to molecular length scales. So, the tangential components of the velocity must match. This is the no-slip boundary condition, and is applicable for all viscous flows. For inviscid flows, we had the normal components matching with $\mathbf{u} \cdot \mathbf{n} = \mathbf{U} \cdot \mathbf{n}$. For viscous flows, even the tangential components match.

3.2.4 Examples

Example 3.1 – Poiseuille flow in a channel



Consider steady flow in a 2D channel driven by a pressure gradient. We have

$$\begin{cases} \partial p / \partial y = 0 \\ \partial p / \partial x = \mu \partial^2 u / \partial y^2 \end{cases} \quad \text{with } u(0) = u(h) = 0.$$

Since $\partial p / \partial x$ depends only on x and $\partial^2 u / \partial y^2$ depends only on y , we must have $\partial p / \partial x$ being constant. Let $\partial p / \partial x = -G = -(p_1 - p_0)/h$. The solution is then $u(y) = Gy(h - y)/(2\mu)$. The flow rate is

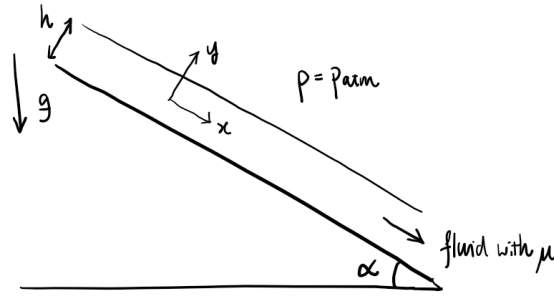
$$q = \int_0^h u \, dy = \frac{(p_1 - p_0)h^3}{12\mu L}.$$

Consider the force on the entire fluid

$$\begin{aligned} p_1 h - p_0 h + \tau_{\text{top}} L - \tau_{\text{bottom}} L &= p_1 h - p_0 h + \mu \frac{\partial u}{\partial y}(h) L - \mu \frac{\partial u}{\partial y}(0) L \\ &= (p_1 - p_0)h + (Gh)L \\ &= 0 \end{aligned}$$

as expected.

Example 3.2 – Viscous flow driven down an incline by gravity



Assume steady flow, so $\mathbf{u} = u(y)\mathbf{e}_x$. Assume that there is no shear stress exerted by air, since $\mu_{\text{air}} \ll \mu_{\text{water}}$. The body force is $\mathbf{f} = \rho\mathbf{g} = (\rho g \sin \alpha, -\rho g \cos \alpha)$. In the y -direction,

$$0 = -\frac{\partial p}{\partial y} - \rho g \cos \alpha \quad \text{with } p(y = h) = p_{\text{atm}}$$

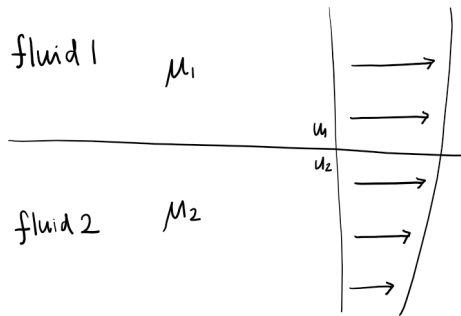
$$p = p_{\text{atm}} + \rho g \cos \alpha (h - y).$$

This resembles hydrostatic balance, only tilted at the angle. Note that $\partial p / \partial x = 0$ since h is constant. Also, $\mu(\partial u / \partial y)(h) = 0$ since there is no shear stress at the top. Hence, in the x -direction,

$$0 = \mu \frac{\partial^2 u}{\partial y^2} + \rho g \sin \alpha \quad \text{with } \begin{cases} u(0) = 0 \\ \mu(\partial u / \partial y)(h) = 0 \end{cases}$$

$$u(y) = \frac{\rho g \sin \alpha}{2\mu} y(2h - y).$$

3.2.5 Boundary conditions at an interface



We have the no-slip condition $u_1 = u_2$ at the interface. Also, we have the continuity of stress. In the normal direction, $p_1 = p_2$. In the tangential direction, $\mu_1 \partial u_1 / \partial y = \mu_2 \partial u_2 / \partial y$ at the interface.

3.3 Unsteady parallel viscous flows and viscous diffusion

A fundamental example is Rayleigh's problem, also known as Stokes' first problem. Consider a semi-infinite ($y > 0$) viscous fluid initially at rest. Assume there is no applied pressure gradient so $\mathbf{f} = \mathbf{0}$. At $t = 0^+$, the plate ($y = 0$) starts to move at constant velocity $(U, 0)$. In the y -direction, $\partial p / \partial y = 0$ so $p \equiv p_0$. In the x -direction,

$$\rho \frac{\partial u}{\partial t} = \mu \frac{\partial^2 u}{\partial y^2}$$

since $\partial p / \partial x = 0$ and $f_x = 0$. Define the kinematic viscosity as

$$\nu = \frac{\mu}{\rho}$$

which has the same dimension as velocity. Together with the no-slip boundary condition, we have

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \quad \text{with} \quad \begin{cases} u(y=0, t) = U \\ u(y \rightarrow \infty, t) = 0. \end{cases}$$

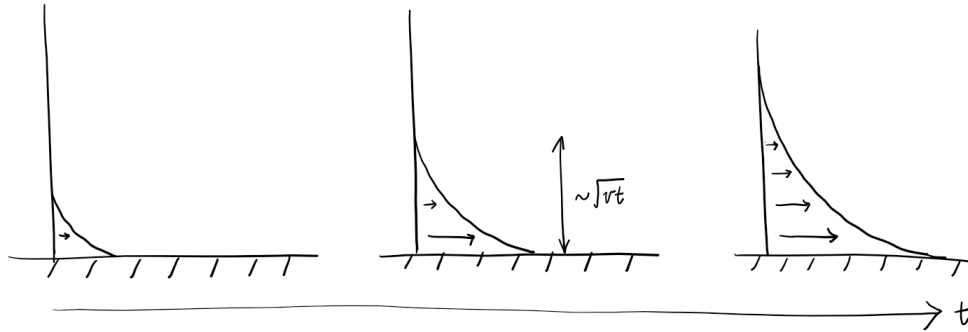
Note that if $\mu = 0$, then there is no friction and no flow. ν represents the diffusivity for momentum, and also for vorticity as we will see later. In IB Methods, we solved the diffusion equation using Fourier methods and separation of variables. Here, we will use dimensional analysis. We know $u(y, t) = g(y, t, U, \nu)$ is linear in U . So, $u(y, t) = U f(y, t, \nu)$. Note that $u/U = f(y, t, \nu)$ is dimensionless. To cancel out the dimensions, we must have

$$\frac{u}{U} = f\left(\frac{y}{\sqrt{\nu t}}\right).$$

Define the similarity variable $\eta = y / \sqrt{\nu t}$. Substituting into the diffusion equation,

$$\begin{aligned} -\frac{1}{2} \frac{\eta}{t} U f' &= \frac{\nu}{\nu t} U f'' & \text{with} \quad \begin{cases} f(0) = 1 \\ f(\infty) = 0 \end{cases} \\ \Rightarrow u(y, t) &= U \operatorname{erfc}\left(\frac{y}{2\sqrt{\nu t}}\right) \end{aligned}$$

where $\operatorname{erfc}(x) = (2/\sqrt{\pi}) \int_x^\infty e^{-t^2} dt$.



Fluid	ρ in kg m^{-3}	μ in Pa s	ν in $\text{m}^2 \text{s}^{-1}$
Water	10^3	10^{-3}	10^{-6}
Air	1	2×10^{-5}	2×10^{-5}
Golden syrup	1.4×10^3	200	0.14

Observe that $\nu_{\text{air}} = 20\nu_{\text{water}}$ so motion spreads faster in air than water. The amount of viscous stress in Rayleigh's problem is

$$\tau = \mu \frac{\partial u}{\partial y}(y=0) = -\frac{\mu U}{\sqrt{\pi \nu t}}.$$

Note that $\mu/\sqrt{\nu}|_{\text{water}} \approx 200 \mu/\sqrt{\nu}|_{\text{air}}$ so it takes 200 times more force to move an object in water than in air.

Example 3.3

Now we add a boundary at $y = h$. Dimensional analysis is no longer possible. When $\sqrt{\nu t} \ll h$, the solution is approximately that in Rayleigh's problem. When t is large, the solution approaches that of Couette flow. The characteristic time scale for the diffusion of momentum across the cell is $t \sim h^2/\nu$. If

$h = 10$ cm, we have $t \sim 0.07$ s for golden syrup, but $t \sim 3$ hrs for water.

Stokes' second problem is similar to the first but with an oscillatory plate $U = \cos(\omega t)$, which is on example sheet 2. During time scales $\sim 1/\omega$, velocity variations can diffuse across a distance $\sim \delta = \sqrt{\nu/\omega}$. If we add another plate at $y = h$, we have two length scales h and δ . The Stokes number is defined as

$$S = \frac{\omega h^2}{\nu} = \frac{h^2}{\delta^2}.$$

On example sheet 3, we will see that the inviscid solution for $\mu = 0$ is not uniformly the same as the limit of small viscosity.

3.4 The Navier-Stokes equations

The following is non-examinable. In Part II, we will derive the most general momentum equation for a Newtonian fluid, namely the Navier-Stokes equations.

$$\nabla \cdot \mathbf{u} = 0 \qquad \rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f} + \underbrace{\mu \nabla^2 \mathbf{u}}_{\text{viscous term}}$$

The vorticity then satisfies

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \underbrace{\nu \nabla^2 \boldsymbol{\omega}}_{\text{viscous term}}.$$

3.5 The Reynolds number

The following is non-examinable. Consider flow with characteristic length scale L and velocity U .

$$\rho \frac{D\mathbf{u}}{Dt} \sim \rho \frac{U^2}{L} \qquad \mu \nabla^2 \mathbf{u} \sim \mu \frac{U}{L^2}$$

Their ratio is Reynolds number

$$\text{Re} = \rho \frac{UL}{\mu} = \frac{UL}{\nu}.$$

For example, $\text{Re} \sim 10^4$ for pouring tea, $\text{Re} \sim 10^{-4}$ for pouring syrup, $\text{Re} \sim 10^6$ for a swimming human. If $\text{Re} \ll 1$, we can safely ignore the left hand side of the Navier-Stokes equation, which corresponds to a quasi-steady limit. If $\text{Re} \gg 1$, the Euler equation is valid almost everywhere except in the thin boundary layers near rigid bodies.

4 Inviscid irrotational flow or potential flow

4.1 Velocity potential

In Chapter 2, we saw that if $\boldsymbol{\omega} = \mathbf{0}$ at $t = 0$, then $\boldsymbol{\omega} = \mathbf{0}$ for all t . That is, irrotational flow remains irrotational. Since $\boldsymbol{\omega} = \nabla \times \mathbf{u} = \mathbf{0}$, we can write $\mathbf{u} = \nabla \phi$ where $\phi = \phi(\mathbf{x}, t)$ is the velocity potential. Note there is no negative sign unlike $\mathbf{f} = -\nabla \chi$. If the flow is incompressible, $\nabla \cdot \mathbf{u} = 0$ implies Laplace's equation

$$\nabla^2 \phi = 0.$$

The kinematic boundary condition $\mathbf{u} \cdot \mathbf{n} = \mathbf{U} \cdot \mathbf{n}$ implies a Neumann boundary condition

$$\frac{\partial \phi}{\partial n} = \mathbf{U} \cdot \mathbf{n}.$$

Hence, solving the Euler equation for irrotational incompressible flow reduces to the more familiar and linear Laplace's equation with Neumann boundary conditions. We first solve Laplace's equation to get $\mathbf{u}$, then use Euler to get the pressure p .

4.2 Simple examples

For simple geometries, we can build solutions using separable solutions to $\nabla^2\phi = 0$ in suitable coordinate systems.

Example 4.1 – Cartesian

If we have uniform flow in a straight pipe, then $\mathbf{u} = \mathbf{U}$ so $\phi = \mathbf{U} \cdot \mathbf{x} = U_x x + U_y y + U_z z$. If ϕ is of the form $\cos kx$, then the flow is periodic in x .

Example 4.2 – Spherical

Recall the general axisymmetric solution to Laplace's equation

$$\phi = \sum_{n \geq 0} (A_n r^n + B_n r^{-1-n}) P_n(\cos \theta)$$

where P_n are the Legendre polynomials. Usually, we only need a small number of modes.

If $\phi = B/r$, then $\mathbf{u} = \nabla\phi = -B/r^2 \mathbf{e}_r$. The volume flow through a sphere of radius r is

$$\int \mathbf{u} \cdot \mathbf{n} dS = -\frac{B}{r^2} (4\pi r^2) = -4\pi B.$$

Defining $q = -4\pi B$, we have $\phi = -q/(4\pi r)$. There is a source at the origin if $q > 0$, and a sink if $q < 0$.

If $\phi = Ur \cos \theta = Uz$, then $\mathbf{u} = \nabla\phi = U\mathbf{z}$, which is uniform flow.

If $\phi = A \cos \theta / r^2$, then it describes a potential dipole, where a source and a sink are close to each other.

Example 4.3 – Cylindrical

The general solution to $\nabla^2\phi = 0$ in 2D polar coordinates is

$$\phi = A_0 \ln r + B_0 \theta + \sum_{n=1}^{\infty} (A_n r^n + B_n r^{-n}) (C_n \cos(n\theta) + D_n \sin(n\theta)).$$

Again, we usually only need a small number of modes.

If $\phi = (q/(2\pi)) \ln r$, then we have radial flow $\mathbf{u} = (q/(2\pi r)) \mathbf{e}_r$. This is a 2D point source or sink. The flow rate across any circle centred at the origin is q .

If $\phi = (\kappa/(2\pi)) \theta$, then we have $\mathbf{u} = (\kappa/(2\pi r)) \mathbf{e}_\theta$. The circulation around any circle centred at the origin is κ . We say that there is a point vortex of strength κ at the origin.

If $\phi = Ur \cos \theta$, then we have uniform flow.

If $\phi = A \cos \theta / r$, then we have a 2D dipole.

4.3 Uniform flow past a sphere or cylinder

4.3.1 Uniform flow past a sphere

$$\begin{cases} \nabla^2 \phi = 0 \\ \mathbf{u} \rightarrow U \mathbf{e}_z \text{ as } r \rightarrow \infty \\ \mathbf{u} \cdot \mathbf{n} = \mathbf{U} \cdot \mathbf{n} = 0 \end{cases} \implies \begin{cases} \nabla^2 \phi = 0 \\ \phi \rightarrow Ur \cos \theta \text{ as } r \rightarrow \infty \\ \partial \phi / \partial r = 0 \text{ at } r = a \end{cases}$$

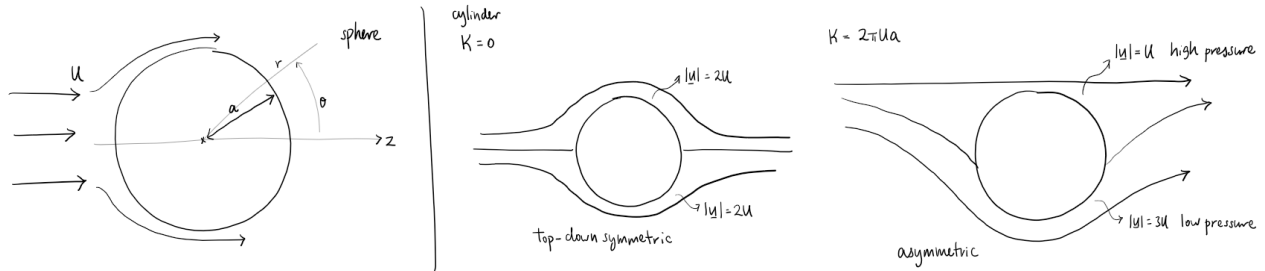
Assume axisymmetric flow. Recall the general solution to Laplace's equation, we had $\cos \theta = P_1(\cos \theta)$. Since $\phi \rightarrow Ur \cos \theta$, we try a solution with only the $n = 1$ term. So, we try the ansatz $\phi = (Ar + B/r^2) \cos \theta$. $\phi \rightarrow Ur \cos \theta$ gives $U = A$.

$$\frac{\partial \phi}{\partial r} = \left(U - \frac{2B}{r^3} \right) \cos \theta$$

so $2B = Ua^3$. This gives the solution

$$\phi = U \cos \theta \left(r + \frac{a^3}{2r^2} \right) \quad \mathbf{u} = \left(U \cos \theta \left(1 - \frac{a^3}{r^3} \right), -U \sin \theta \left(1 + \frac{a^3}{2r^3} \right), 0 \right).$$

We see that ϕ describes a uniform flow and a dipole.



4.3.2 Uniform flow past a cylinder with circulation κ

$$\begin{cases} \nabla^2 \phi = 0 \\ \phi \rightarrow Ur \cos \theta \text{ as } r \rightarrow \infty \\ \partial \phi / \partial r = 0 \text{ at } r = a \\ \oint \mathbf{u} \cdot d\mathbf{l} = [\phi]_{\theta=0}^{2\pi} = \kappa \end{cases}$$

Working in 2D polar coordinates, the solution is

$$\phi = \frac{\kappa}{2\pi} \theta + U \cos \theta \left(r + \frac{a^2}{r} \right) \quad \mathbf{u} = \left(U \cos \theta \left(1 - \frac{a^2}{r^2} \right), -U \sin \theta \left(1 + \frac{a^2}{r^2} \right) + \frac{\kappa}{2\pi r} \right).$$

The $(\kappa/(2\pi))\theta$ term in ϕ describes a singular vortex inside the cylinder.

4.4 Equation for pressure in potential flow with $\mathbf{f} = -\nabla \chi$

To solve for the flow $\mathbf{u}$, we used $\nabla^2 \phi = 0$ and boundary conditions. To get the pressure p , we use the Euler equation $\rho(D\mathbf{u}/Dt) = -\nabla p + \mathbf{f}$, but this is nonlinear. We will show that if $\mathbf{f} = -\nabla \chi$ and ρ is constant, then

there is an explicit equation for p .

$$\begin{aligned}\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) &= -\nabla p + \mathbf{f} \\ \rho \left(\frac{\partial}{\partial t}(\nabla \phi) + \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times \boldsymbol{\omega} \right) &= -\nabla p - \nabla \chi \\ \nabla \left(\rho \frac{\partial \phi}{\partial t} \right) + \nabla \left(\frac{1}{2} \rho |\nabla \phi|^2 \right) &= -\nabla p - \nabla \chi\end{aligned}$$

since we assumed irrotational flow $\boldsymbol{\omega} = \mathbf{0}$ in this chapter. Hence,

4.1 – Unsteady Bernoulli equation

$$\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho |\nabla \phi|^2 + p + \chi = f(t)$$

This is exact. Steady Bernoulli from Chapter 2 is a consequence of Euler's equation, and assumed steady flow and allowed vorticity. However, unsteady Bernoulli is equivalent to Euler's equation, and assumed zero vorticity and allowed unsteady flow. Both required a conservative force $\mathbf{f} = -\nabla \chi$.

4.5 Forces on a translating sphere or cylinder

4.5.1 Translating sphere

Consider a sphere of radius a moving with speed U through some stationary fluid. In the reference frame of the sphere, there is a steady flow as in Section 4.3.1. The force on the sphere is $-\int p \mathbf{n} dS$. On the sphere, we have $u_r = 0$ and $u_\theta = -(3/2)U \sin \theta$. By either steady or unsteady Bernoulli, we get

$$\begin{aligned}\frac{1}{2} \rho \left[-\frac{3}{2} U \sin \theta \right]^2 + p &= \frac{1}{2} \rho U^2 + p_\infty \\ p - p_\infty &= \frac{1}{2} \rho U^2 \left[1 - \frac{9}{4} \sin^2 \theta \right].\end{aligned}$$

We see that the pressure distribution is symmetric, so the net force on the sphere is zero. This is known as d'Alembert's paradox. In fact, there is no drag (component of the hydrodynamic force parallel to the motion) on any steadily moving rigid body in an unbounded potential flow. In real life, there is flow separation and this would not hold. However, potential flow approximation is valid for bubbles, rapid acceleration of rigid particles, or for small amplitude oscillations.

4.5.2 Steady translation of a cylinder with circulation κ

We again work in the frame of the cylinder. On the sphere, $u_r = 0$ and $u_\theta = -2U \sin \theta + \kappa/(2\pi a)$. Bernoulli gives

$$\frac{1}{2} \rho \left[-2U \sin \theta + \frac{\kappa}{2\pi a} \right]^2 + p = \frac{1}{2} \rho U^2 + p_\infty.$$

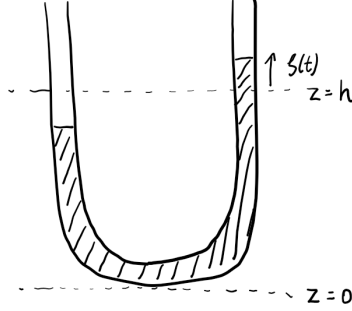
The force per unit length on the cylinder is

$$\begin{aligned}\mathbf{F} &= -\int p \mathbf{n} dS \\ &= -\int_0^{2\pi} \left(\frac{1}{2} \rho U^2 + p_\infty - \frac{1}{2} \rho \left(4U^2 \sin^2 \theta - \frac{2U\kappa}{\pi a} \sin \theta + \frac{\kappa^2}{4\pi^2 a^2} \right) \right) (\cos \theta, \sin \theta) a d\theta \\ &= (0, -\rho U \kappa).\end{aligned}$$

The drag is still zero, but there is a force perpendicular to the motion. We call this the lift force. If $\kappa > 0$, then $F_y < 0$. If $\kappa < 0$, then $F_y > 0$. On example sheet 3, we will show that uniform flow past any 2D body with circulation κ induces lift $-\rho U \kappa$.

4.6 Unsteady potential flows

4.6.1 Free oscillations in a tube



Assume we start with irrotational flow so it remains irrotational. Assume a uniform circular cross section with radius a . The flow is uniform with $u = \dot{\zeta}$. In the right tube, $\phi = \dot{\zeta}z$ so $[\partial\phi/\partial t]_{h+\zeta} = \ddot{\zeta}(h + \zeta)$. In the left tube, $\phi = -\dot{\zeta}z$ so $[\partial\phi/\partial t]_{h-\zeta} = -\ddot{\zeta}(h - \zeta)$. By considering the fluid surface of the right tube and that of the left tube, unsteady Bernoulli gives

$$\rho\ddot{\zeta}(h + \zeta) + \frac{1}{2}\rho\dot{\zeta}^2 + p_{\text{atm}} + \rho g(h + \zeta) = -\rho\ddot{\zeta}(h - \zeta) + \frac{1}{2}\rho\dot{\zeta}^2 + p_{\text{atm}} + \rho g(h - \zeta)$$

$$\ddot{\zeta} + \frac{g}{h}\zeta = 0$$

so we have oscillations with frequency $\omega = \sqrt{g/h}$.

4.6.2 Oscillations of a bubble

Consider incompressible flow of a liquid outside a compressible 3D bubble of a gas of radius $a(t)$. Assume spherical symmetry so we have a radial flow. By mass conservation, $u \sim 1/r^2$. The boundary condition gives $u|_{r=a} = \dot{a}$. Thus,

$$\mathbf{u} = \dot{a} \frac{a^2}{r^2} \mathbf{e}_r$$

$$\phi = -a^2 \frac{\dot{a}}{r}$$

$$\frac{\partial\phi}{\partial t} = -\frac{2\dot{a}^2 a + a^2 \ddot{a}}{r}$$

$$\rho \left[\frac{\partial\phi}{\partial t} \right]_{r=a} = -\rho(2\dot{a}^2 + a\ddot{a}).$$

Ignoring gravity, unsteady Bernoulli gives

$$\rho \left[\frac{\partial\phi}{\partial t} \right]_{r=a} + \frac{1}{2}\rho(u(a))^2 + p(a, t) = \rho \left[\frac{\partial\phi}{\partial t} \right]_{r=\infty} + \frac{1}{2}\rho(u(\infty))^2 + p_\infty$$

$$-\frac{3}{2}\rho\dot{a}^2 - \rho a\ddot{a} = p_\infty - p(a, t).$$

If the oscillations are small, we can write $a = a_0 + \eta$ with $\eta \ll a_0$. This gives $\rho a_0 \dot{\eta} = p - p_\infty$. How does $p = p_{\text{gas}}$ change when a changes? We will assume that PV^γ is constant. Since $V \sim a^3$, we have

$$\frac{\delta P}{P} = -\gamma \frac{\delta V}{V} = -3\gamma \frac{\delta a}{a} \implies \frac{\delta P}{p_\infty} = -3\gamma \frac{\eta}{a_0}.$$

Hence,

$$\rho a_0 \ddot{\eta} = -\frac{3\gamma p_\infty}{a_0} \eta$$

and the oscillations have frequency $\omega = [3\gamma p_\infty / (\rho a_0^2)]^{1/2}$. An air bubble in water has $\omega \approx 2 \times 10^4$ Hz.

4.6.3 Accelerating sphere

Recall Section 4.3.1, but we work in the lab frame since the sphere does not provide an inertial frame. We had

$$\phi = \frac{Ua^3 \cos \theta}{2r^2} \implies \phi = -\frac{a^3 \mathbf{U}(t) \cdot \mathbf{x}}{2r^3}.$$

When $\mathbf{U}(t)$ is steady, we had the pressure on the sphere $p = 0$. When $\mathbf{U}(t)$ is changing,

$$\rho \frac{\partial \phi}{\partial t} = -\rho a^3 \frac{\dot{\mathbf{U}} \cdot \mathbf{x}}{2r^3} \implies \rho \left[\frac{\partial \phi}{\partial t} \right]_{r=a} = -\frac{1}{2} \rho \dot{\mathbf{U}} \cdot \mathbf{x}.$$

By unsteady Bernoulli, the pressure = p_{steady} + new unsteady term, so

$$\begin{aligned} p &= p_\infty + \frac{1}{2} \rho U^2 \left(1 - \frac{9}{4} \sin^2 \theta \right) + \frac{1}{2} \rho \dot{\mathbf{U}} \cdot \mathbf{x} \quad \text{so } \mathbf{F} = - \int_S p \mathbf{n} dS \\ &= - \int_S \frac{1}{2} \rho (\dot{\mathbf{U}} \cdot \mathbf{x}) \mathbf{n} dS \quad \text{since } p_{\text{steady}} \text{ vanishes} \\ &= -\frac{1}{2} \rho \dot{\mathbf{U}} \cdot \int_S \mathbf{x} \mathbf{n} dS \\ &= -\frac{1}{2} \rho \dot{\mathbf{U}} \cdot a \int_S \mathbf{n} \mathbf{n} dS \quad \text{since } \mathbf{x} = a \mathbf{n} \text{ on the surface} \\ &= -\frac{1}{2} \rho \dot{\mathbf{U}} a \frac{4\pi}{3} a^2 \\ &= -\frac{1}{2} \rho V \underbrace{\dot{\mathbf{U}}}_{m^*} \end{aligned}$$

where m^* is known as the added mass. In general, $m^* = \alpha \rho V$ with α being the added mass coefficient. Why is called an added mass? Consider a sphere of mass m subject to an external force $\mathbf{F}_{\text{ext}}$. By Newton's second law, $m\dot{\mathbf{U}} = \mathbf{F}_{\text{ext}} - m^*\dot{\mathbf{U}}$ so

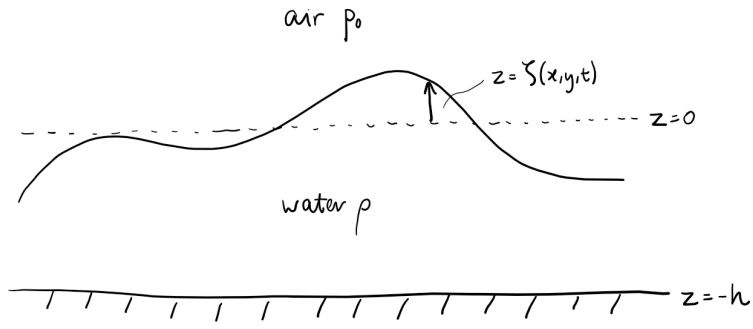
$$(m + m^*)\dot{\mathbf{U}} = \mathbf{F}_{\text{ext}}$$

and the body appears heavier.

5 Geophysical flows

5.1 Water waves

We can apply what we derived in Chapter 4 to ocean waves, ships waves, river flows, etc.



5.1.1 Governing equations

Assume that the flow is inviscid and that the motion starts from rest. So, the flow remains irrotational $\omega = 0$. Setting $\mathbf{u} = \nabla\phi$, we get

$$\nabla^2\phi = 0 \quad \text{in } -h < z < \zeta.$$

The kinematic boundary condition at the bottom requires $\mathbf{n} \cdot \nabla\phi = 0$ so

$$\frac{\partial\phi}{\partial z} = 0 \quad \text{at } z = -h.$$

As derived in Chapter 1, we need

$$\frac{\partial\zeta}{\partial t} + u\frac{\partial\zeta}{\partial x} + v\frac{\partial\zeta}{\partial y} = w \quad \text{at } z = \zeta$$

where $\mathbf{u} = (u, v, w)$. The dynamic boundary condition requires $p_{\text{water}} = p_0$ at $z = \zeta$. Applying unsteady Bernoulli at the surface, we get

$$\rho\frac{\partial\phi}{\partial t} + \frac{1}{2}\rho|\nabla\phi|^2 + p_0 + \rho gz = f(t) \quad \text{at } z = \zeta.$$

This is a nonlinear problem, evaluated at the unknown interfacial position.

5.1.2 Linear water waves

Assume small amplitude, so $\zeta \ll h$ and $|\partial\zeta/\partial x|, |\partial\zeta/\partial y| \ll 1$. We linearise the equations about its rest state. We ignore the nonlinear terms and do Taylor expansions around $z = 0$ for all quantities evaluated at $z = \zeta$. This is known as domain perturbation expansion. For example,

$$\left[\frac{\partial\phi}{\partial z}\right]_{z=\zeta} = \left[\frac{\partial\phi}{\partial z}\right]_{z=0} + \zeta \left[\frac{\partial^2\phi}{\partial z^2}\right]_{z=0} + \dots$$

The resulting linearised equations are

$$\begin{aligned} \nabla^2\phi &= 0 & \text{in } -h < z < 0 \\ \frac{\partial\phi}{\partial z} &= 0 & \text{at } z = -h \\ \frac{\partial\zeta}{\partial t} &= \frac{\partial\phi}{\partial z} & \text{at } z = 0 \\ \rho\frac{\partial\phi}{\partial t} + \rho g\zeta &= F(t) & \text{at } z = 0. \end{aligned}$$

What is the solution? In IB Methods, we looked for separable solutions $\phi = A(z)B(x)C(y)T(t)$, or did Fourier transforms when in an infinite domain. In 2D, we end up looking for solutions of the form

$$\begin{cases} \zeta = \zeta_0 \exp[i(kx - \omega t)] \\ \phi = \Phi(z) \exp[i(kx - \omega t)] \end{cases}$$

where the real parts are taken. The general solution is a superposition of an infinite number of Fourier modes. (1) implies $\Phi'' - k^2\Phi = 0$ so $\Phi \sim e^{\pm kz}$. It is convenient to write this as $\Phi = \sinh(k(z+h)) + B \cosh(k(z+h))$. (2) gives $B = 0$ so $\Phi = A \cosh(k(z+h))$. The right hand side of (4) does not depend on x so $F(t) = 0$. Combining with (3) gives

$$\begin{cases} -i\omega\zeta_0 = Ak \sinh(kh) \\ -i\omega A \rho \cosh(kh) + \rho g \zeta_0 = 0. \end{cases}$$

This linear homogenous equation has non-zero solution if and only if

$$\begin{vmatrix} -i\omega & -k \sinh(kh) \\ \rho g & -i\omega \rho \cosh(kh) \end{vmatrix} = \rho[-\omega^2 \cosh(kh) + gk \sinh(kh)] = 0.$$

Hence, the solutions must obey the dispersion relationship $\omega^2 = gk \tanh(kh)$. The phase speed is

$$c = \frac{\omega}{k} = \sqrt{\frac{g \tanh(kh)}{h}}.$$

We see that water waves are dispersive. Waves of different wavelength travel at different speeds.

In deep water with $kh \gg 1$, we have $\omega = \sqrt{gk}$ and $c = \sqrt{g/k}$ which is independent of h . For example, in storm waves in the Atlantic ocean, we observe period ~ 15 s and $\omega \sim 0.4$ rad s⁻¹. This is indeed deep water since $60 \text{ m} \sim k^{-1} \ll \text{ocean depth} \sim 4 \text{ km}$. We get $c = \omega/k \sim 2000 \text{ km/day}$. In shallow water with $kh \ll 1$, we have $\omega = \sqrt{gh}k$ and $c = \sqrt{gh}$ which is independent of k , so waves are approximately not dispersive. In a tsunami, $k^{-1} \sim 500 \text{ km}$ so $c \sim 200 \text{ m s}^{-1}$.

We can also calculate the velocity. $\phi = -i\omega\zeta_0 \cosh(k(z+h)) / (k \sinh(kh)) e^{i(kx - \omega t)}$ so

$$\mathbf{u} = \frac{\zeta_0 \omega}{\sinh(kh)} e^{i(kx - \omega t)} (\cosh(k(z+h)), -i \sinh(k(z+h))).$$

In the deep water limit, the trajectories of the fluid particles are approximately circles. In the shallow water limit, we have large longitudinal motion in the x -direction, but very small motion in the z -direction.

5.1.3 Standing waves in a container

Consider water in a semi-infinite rectangular container with $0 \leq x \leq a$ and $0 \leq y \leq b$. Let the mean water level be $z = 0$. We again look for linearised waves so

$$\begin{cases} \nabla^2 \phi = 0 & \text{in } z < 0 \\ \partial \phi / \partial n = 0 & \text{at the side walls} \\ \nabla \phi \rightarrow \mathbf{0} & \text{as } z \rightarrow -\infty \\ \partial \zeta / \partial t = \partial \phi / \partial z & \text{at } z = 0 \\ \rho(\partial \phi / \partial t) + \rho g \zeta \text{ is independent of } x, y. \end{cases}$$

Separation of variables give

$$\phi = A \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{kz} e^{-i\omega t}$$

where m and n are integers. We have

$$k^2 = \frac{m^2 \pi^2}{a^2} + \frac{n^2 \pi^2}{b^2} = k_{mn}^2$$

and $\omega^2 = gk$. This is different with the infinite case as k is quantised. The modes are called standing waves or normal modes.

5.1.4 Rayleigh-Taylor instability

Consider a container upside-down, with water at the top and air at the bottom. It is obviously unstable, with g replaced by $-g$. Thus, $\omega^2 = -gk$ so $\omega = \pm i\sqrt{gk}$. This gives exponential growth $\exp(i\omega t) = \exp(\pm\sqrt{gk}t)$.

5.2 Rotating fluid dynamics

We live on a rotating planet with $\Omega = 2\pi/86400 \sim 10^{-4} \text{ s}^{-1}$. Earth's rotation has a huge effect on the motion of our ocean and atmosphere. Moreover, the ocean ($\sim 4 \text{ km}$) and the atmosphere ($\sim 10 \text{ km}$) are thin relative to the Earth's radius ($\sim 6000 \text{ km}$). The rotational speed of the Earth at the equator is $U = \Omega R \sim 500 \text{ m s}^{-1}$, which is much larger than any wave or wind speed.

5.2.1 Euler equations in a rotating frame

Assume ρ is constant and $\nabla \cdot \mathbf{u} = 0$. This is very good approximation for the ocean, and a good approximation for the atmosphere. The acceleration of a fluid particle in a rotating frame becomes

$$\frac{D\mathbf{u}}{Dt} \mapsto \frac{D\mathbf{u}}{Dt} + \underbrace{2\boldsymbol{\Omega} \times \mathbf{u}}_{\text{Coriolis}} + \underbrace{\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})}_{\text{centrifugal}}.$$

Note that $\rho\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}) = \nabla[\rho|\boldsymbol{\Omega} \times \mathbf{x}|^2/2]$ so we can absorb the centrifugal term into ∇p . Also, we can neglect the centrifugal term since $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})/g \sim 1/300 \ll 1$. Hence, the Euler equation in a rotating frame is

$$\rho \frac{D\mathbf{u}}{Dt} + 2\rho\boldsymbol{\Omega} \times \mathbf{u} = -\nabla p + \rho\mathbf{g}.$$

Taking the curl, we get

$$\frac{D\boldsymbol{\omega}}{Dt} = ((\boldsymbol{\omega} + 2\boldsymbol{\Omega}) \cdot \nabla)\mathbf{u}.$$

The vortex stretching now includes two terms, the relative vorticity $\boldsymbol{\omega}$ and the planetary vorticity $\boldsymbol{\Omega}$.

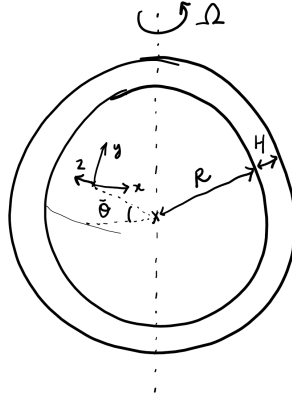
5.2.2 Rossby number

How important is the Coriolis effect? Consider flow with characteristic velocity U and characteristic length L . $\rho D\mathbf{u}/Dt \sim \rho U/(L/U) = \rho U^2/L$ and $2\rho\boldsymbol{\Omega} \times \mathbf{u} \sim \rho\Omega U$ so

$$\frac{\rho \frac{D\mathbf{u}}{Dt}}{2\rho\boldsymbol{\Omega} \times \mathbf{u}} \sim \frac{\rho \frac{U^2}{L}}{\rho\Omega U} = \frac{U}{\Omega L} =: \text{Ro}$$

is the Rossby number, quantifying the relative magnitude of inertia versus rotation. For example, if $\Omega \sim 10^{-4} \text{ s}^{-1}$, $L \sim 1000 \text{ km}$ and $U \sim 10 \text{ m s}^{-1}$, then $\text{Ro} \sim 10^{-1}$. When $\text{Ro} \ll 1$, $\boldsymbol{\omega} \sim U/L \ll \boldsymbol{\Omega}$ so the vorticity is dominated by the planetary component.

5.2.3 Rotating flows in shallow layer



Consider flow over a characteristic length scale L . Assume $L \ll R$ so it is locally flat. Assume $H \ll L$ so it is a shallow layer. Define local Cartesian coordinates with x being east, y being north, and z being up. Let $\bar{\theta}$ be the latitude. Let $\mathbf{u} = (u, v, w)$. Then, $\boldsymbol{\Omega} = \Omega(0, \cos \bar{\theta}, \sin \bar{\theta})$ so

$$2\boldsymbol{\Omega} \times \mathbf{u} = 2\Omega(w \cos \bar{\theta} - v \sin \bar{\theta}, u \sin \bar{\theta}, -u \cos \bar{\theta}).$$

We make three assumptions.

- (i) $u \sim 10 \text{ m s}^{-1}$ and $\Omega u/g \sim 10^{-4} \ll 1$ so we neglect the z component in the Coriolis term.
- (ii) $\nabla \cdot \mathbf{u} = 0$ so $u/L, v/L \sim w/H$ so $w \ll u, v$ since $H \ll L$. Thus, we can neglect the $w \cos \bar{\theta}$ term in the Coriolis x component. This approximation breaks down near the equator.
- (iii) w is small so we can neglect the vertical acceleration $\rho Dw/Dt$. This gives us hydrostatic balance.

Hence, we arrive at the following simplified equations. Let $f = 2\Omega \sin \bar{\theta}$ be the Coriolis parameter.

$$\begin{aligned} \frac{Du}{Dt} - fv &= -\frac{1}{\rho} \frac{\partial p}{\partial x} \\ \frac{Dv}{Dt} + fu &= -\frac{1}{\rho} \frac{\partial p}{\partial y} \\ 0 &= -\frac{\partial p}{\partial z} + \rho g \\ \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} &= 0 \end{aligned}$$

Taking $(\partial/\partial y)(\text{eq. 1}) - (\partial/\partial x)(\text{eq. 2})$ and calling $\xi = \omega_z$, we get

$$\frac{D\xi}{Dt} = (\xi + f) \frac{\partial w}{\partial z} = -(\xi + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)$$

which is the new form of the vorticity equation.

Consider flow in a shallow layer with thickness $h(x, y, t)$. Equation 3 gives

$$p = p_0 + \rho g(h - z) \implies \begin{cases} \frac{\partial p}{\partial x} = \rho g \frac{\partial h}{\partial x} \\ \frac{\partial p}{\partial y} = \rho g \frac{\partial h}{\partial y} \end{cases} \implies \nabla_H p = \rho g \nabla_H h$$

which is independent of z . So, if u and v are independent of z at $t = 0$, they remain so for later t . The subscript

H represents derivatives in the horizontal direction. For mass conservation, consider a cylinder A . We have

$$\begin{aligned}\frac{d}{dt} \int_A \rho h dA + \oint_{\partial A} \rho h \mathbf{u}_H \cdot \mathbf{n} dS &= 0 \\ \frac{\partial h}{\partial t} + \nabla_H \cdot (h \mathbf{u}_H) &= 0\end{aligned}$$

by the divergence theorem. Or equivalently,

$$\frac{Dh}{Dt} + h \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0.$$

5.2.4 Geostrophic balance

When the flow is steady and $Ro \ll 1$, equations 1 and 2 become

$$\begin{cases} -fv = -\frac{1}{\rho} \frac{\partial p}{\partial x} \\ fu = -\frac{1}{\rho} \frac{\partial p}{\partial y} \end{cases} \implies \begin{cases} u = \frac{\partial}{\partial y} [-gh/f] \\ v = -\frac{\partial}{\partial x} [-gh/f] \end{cases}$$

using hydrostatic balance, so the streamfunction is

$$\psi = -\frac{gh}{f} = -\frac{p}{\rho f}.$$

Thus, the fluid flows along the isobars (lines of constant pressure).

5.2.5 Potential vorticity

Recall that

$$\begin{cases} \frac{D}{Dt} (\zeta + f) = \frac{D\zeta}{Dt} = -(\zeta + f) \nabla_H \cdot \mathbf{u}_H \\ \frac{Dh}{Dt} = -h \nabla_H \cdot \mathbf{u}_H \end{cases} \implies \frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = 0.$$

$(\zeta + f)/h$ is known as the potential vorticity, and is conserved.

5.2.6 Linearised evolution

Assume $h = h_0 + \eta(x, y, t)$ with $|\eta| \ll h$. The linearised equations are

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -\frac{1}{\rho} \frac{\partial p}{\partial x} = -g \frac{\partial h}{\partial x} \\ \frac{\partial v}{\partial t} + fu &= -\frac{1}{\rho} \frac{\partial p}{\partial y} = -g \frac{\partial h}{\partial y} \\ \frac{\partial \eta}{\partial t} + h_0 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) &= 0.\end{aligned}$$

Conservation of potential vorticity linearised is

$$\frac{\partial}{\partial t} \left(\frac{\zeta + f}{h_0 + \eta} \right) = 0 \implies \zeta - \eta \frac{f}{h_0} = \text{const.}$$

Taking $h_0((\partial/\partial x)(\text{linearised eq. 1}) + (\partial/\partial y)(\text{1.eq. 2})) - (\partial/\partial t)(\text{1.eq. 3})$, we get

$$\frac{\partial^2 \eta}{\partial t^2} - c^2 \nabla^2 \eta + f^2 \eta = f^2 \eta_0 + h_0 f \zeta_0$$

where $\eta_0 = \eta(t = 0)$, $\xi_0 = \xi(t = 0)$ and $c = \sqrt{gh_0}$ is the shallow-water wave speed.

$$a = \frac{c}{f} = \frac{\sqrt{gh_0}}{f}$$

is known as the Rossby radius of deformation. In the ocean, $a \sim 10$ km at the poles and $a \sim 200$ km near the equator.